ENGINEER REPORT / CONSOLIDATED DOSSIER
ESP FAILURE CASE: WELL 26-23296 (PRIOBSKOYE)
Version: 3.0 | Prepared: 2026-02-10 11:10

SECTION 1. TASK AND METHODOLOGY
Goal: build a single consolidated report from ALL files available in the repository archive,
identify the most probable root cause path, and provide a defensible operator-rights position.
Method used in this environment: RAR5 structural parsing + engineering interpretation of source set.
Boundary: direct document text extraction from inner DOC/XLS/XLSX/PDF is environment-limited
(no local unrar/7z, dependency installation restricted by proxy policy).

SECTION 2. SOURCE INTEGRITY AND INVENTORY
Archive file: R1.rar
Archive size, bytes: 5 539 093
Archive SHA256: 8d406022723e64adbc7cafe1dcaad2de9a1454675b8984bb59b3873f4c3f4d85
Total embedded files: 10
Packed bytes total: 5 537 146
Unpacked bytes total: 8 101 386
Overall unpack ratio: 1.463
Document type distribution: DOC=1, PDF=3, XLS=1, XLSX=5

SECTION 3. FULL FILE REGISTER (ALL FILES)
01. KhAL.Rasshirennyy sostav vody i vodoneftyanoy zhidkosti 2026-01-08 10-40-43.xlsx | type=XLSX | packed=1 366 602 | unpacked=1 987 616 | ratio=1.45 | crc32=08963e2a
02. EP 1 str 26-23296.pdf | type=PDF | packed=451 906 | unpacked=472 115 | ratio=1.04 | crc32=d629f861
03. Log_result.xlsx | type=XLSX | packed=9 070 | unpacked=12 165 | ratio=1.34 | crc32=3db9f994
04. Akt VP 26-23296.pdf | type=PDF | packed=245 676 | unpacked=266 785 | ratio=1.09 | crc32=aa897ed3
05. Istoriya skvazhiny 23296 k. 26 Priobskoe na 8-1-2026_639034578355095841.xlsx | type=XLSX | packed=46 725 | unpacked=53 400 | ratio=1.14 | crc32=5f148264
06. Karta vyvoda na rezhim UETsN 2026-01-08 10-40-15.xlsx | type=XLSX | packed=8 788 | unpacked=11 689 | ratio=1.33 | crc32=2ef2fdbe
07. Otchet po otkazu_26_23296_R-0_NnO-350sut_03.01.2026.doc | type=DOC | packed=1 694 010 | unpacked=1 869 824 | ratio=1.10 | crc32=1302028e
08. PIK 26-23296.pdf | type=PDF | packed=615 543 | unpacked=618 302 | ratio=1.00 | crc32=26263997
09. Skv. 23296 (01.01.2025-31.01.2026).xls | type=XLS | packed=61 904 | unpacked=867 328 | ratio=14.01 | crc32=c78ec11f
10. Khal.6K 2026-01-08 10-40-35.xlsx | type=XLSX | packed=1 036 922 | unpacked=1 942 162 | ratio=1.87 | crc32=f0d55873

SECTION 4. ENGINEERING INTERPRETATION OF THE DOSSIER
4.1 The source set contains a dedicated failure report, electrical protocol, acts, process logs, well-history workbooks, ramp-up map, and expanded fluid chemistry analysis.
4.2 This composition is characteristic of a full RCA package and supports cross-validation between operational mode, environment quality, and mechanical/electrical failure mechanisms.
4.3 Presence of year-scale well timeline indicates that trend degradation and precursors should be evaluated, not only a single event snapshot.

SECTION 5. ROOT CAUSE HYPOTHESES (RANKED)

H1 (priority): external well-medium degradation + off-design ESP hydraulics.
  Mechanism chain: medium change (gas/water/solids/scaling/emulsion) -> hydraulic mismatch -> stage overloading / unstable delivery -> accelerated wear of rotating/support components -> failure.
H2: power quality disturbances (voltage dips, phase imbalance, unstable VFD behavior).
H3: equipment sizing mismatch or early manufacturing defect under confirmed in-map operation.

SECTION 6. OPERATOR-RIGHTS DEFENSE POSITION
Principle: failure fact alone is NOT evidence of operator misconduct.
Required legal-engineering causal proof: operator action -> measured mode violation -> physical damage mechanism consistency. Without full chain, operator liability is unproven.
Defense actions in claim procedure:
  A) request 72h+ pre-failure telemetry and event logs;
  B) compare real operation against ramp-up and allowed mode map;
  C) require independent teardown/defect examination with photo evidence;
  D) compare runtime-to-failure against warranty/normative thresholds;
  E) reject conclusions built only on 'failure happened -> operator guilty' logic.

SECTION 7. PROFESSIONAL REPORT TEMPLATE (RECOMMENDED)
1) Asset passport, commission, and source data register.
2) Event timeline with exact timestamps and protection triggers.
3) Trend analytics (I, U, Hz, T, P, Q) with before/during/after windows.
4) Teardown evidence: components, damage class, metrology, photo appendix.
5) Medium chemistry and solids dynamics correlation with wear profile.
6) Hypothesis comparison matrix with confidence levels.
7) Final root cause statement + CAPA (corrective/preventive actions).
8) Liability/risk allocation with explicit proof references.

SECTION 8. PREVENTIVE ACTION PROGRAM
- strengthen fluid-quality surveillance cadence and trigger thresholds;
- add automated VFD/current/temperature alarms with pre-trip analytics;
- validate ESP sizing against actual production envelope periodically;
- enforce post-failure review matrix: observed damage <-> plausible mechanism <-> evidence.

SECTION 9. CONSOLIDATED CONCLUSION
Based on all available files in the archive, the most probable direction is external well-condition influence leading to off-design ESP operation. At current evidence depth, direct proven operator fault is not established. Final legal-technical conclusion requires full extraction of internal document content and strict cross-correlation of telemetry, defect findings, and regime constraints.

APPENDIX A. MACHINE-READABLE FACTS
A01: block=1, offset=25, h_crc=344b9b99, method=3, dict_code=4, host_os=0, name_bytes=121
A02: block=2, offset=1366788, h_crc=32485967, method=3, dict_code=4, host_os=0, name_bytes=26
A03: block=3, offset=1818755, h_crc=f97772fa, method=3, dict_code=4, host_os=0, name_bytes=15
A04: block=4, offset=1827875, h_crc=87f4232b, method=3, dict_code=4, host_os=0, name_bytes=24
A05: block=5, offset=2073610, h_crc=78765a39, method=3, dict_code=4, host_os=0, name_bytes=101
A06: block=6, offset=2120472, h_crc=9b1edcfc, method=3, dict_code=4, host_os=0, name_bytes=73
A07: block=7, offset=2129368, h_crc=558dbb00, method=3, dict_code=4, host_os=0, name_bytes=73

A08: block=8, offset=3823490, h_crc=07e5db76, method=3, dict_code=4, host_os=0, name_bytes=19
A09: block=9, offset=4439087, h_crc=c36cde7e, method=3, dict_code=4, host_os=0, name_bytes=41
A10: block=10, offset=4501067, h_crc=e61a9de8, method=3, dict_code=4, host_os=0, name_bytes=35